LESSON PLAN

GAS TURBINE ENGINE (16)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Radiographic inspection, use, applicability inspection principle of radiography, X-ray gamma ray, shadow for motion, types of recording, types of test specimen safety precaution while using x-ray
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Radiographic inspection, use, applicability inspection principle of radiography, X-ray gamma ray, shadow for motion, types of recording, types of test specimen safety precaution while using x-ray
5. Specific Objectives	16.1.2.1 Trainee will be able to explain principle of radiographic inspection.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of radiographic film interpretation using sample films; Safety briefing on X-ray source handling and shielding protocols; Instructor supervision of safety drill execution.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of non-destructive testing in engine maintenance?
(b) Name two common methods used for inspecting internal engine components.
(c) Define the term 'radiographic inspection' in one sentence.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principle of radiographic inspection • Interaction of X-ray and Gamma ray with matter • Shadow formation and motion blur effects	L & Ds	PPT, IB	15 min
(b)	• Types of recording media • Applicability of radiography in gas turbine engines • Types of test specimens used	L & Ds	PPT, PDF	20 min
(c)	• Safety precautions during X-ray operations • Use of shielding and dosimetry • Emergency procedures for radiation exposure	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the density of a material affects the shadow on a radiographic film.
(b) List three safety precautions required when operating an X-ray machine.
(c) Differentiate between X-ray and Gamma ray sources in terms of applicability.

12. Home Assignment and Follow Up	Prepare a brief report comparing the advantages and limitations of radiographic inspection versus ultrasonic testing for detecting cracks in turbine blades, referencing the principles covered in this lesson and preparing for the introduction to neutron radiography in LP 06.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Inspection of neutron radiography introduction, different methods principle and various applications	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min